



KiCad Libraries Management

Personal • Global • Project-specific • Git-backed corporate libraries

A scalable approach to KiCad's component libraries

Audience: Teams using KiCad in production

Hierarchy • Sources • Git workflow • Corporate libs • Versioning



KiCad Library Hierarchy

- ▶ Built-in libraries: shipped with KiCad (KiCad maintained)
- ▶ Global libraries: configured at user level (Tools → Manage Symbol/Footprint Libraries)
- ▶ Project libraries: specific to one project (in project folder)
- ▶ Search order: project → global → built-in (project libs take priority)
- ▶ All libraries listed in: `~/.config/kicad/<version>/sym-lib-table + fp-lib-table`
- ▶ Project-specific libs: `<project>/sym-lib-table + fp-lib-table` (override global)
- ▶ Best practice: use project libs for any non-standard component (vendor-specific)

Library Sources

KiCad built-in

5000+ symbols + footprints
Well-maintained, free.

SnapEDA

5M+ KiCad parts.
Free (limited), paid bulk.

Ultra Librarian

Web-based, vendor-supplied.
Free download per part.

Component Search Engine

Vendor parts library.
Free, fast download.

Vendor direct

ST, Espressif, Nordic
often ship KiCad libs.

Custom

Your own + corporate libs.
Full control.

Recommended Library Structure

```
# Project structure for a typical KiCad project
my_pcb_project/
├── my_project.kicad_pro      # Project file
├── my_project.kicad_sch     # Schematic
├── my_project.kicad_pcb     # PCB
├── sym-lib-table            # Project-specific symbol lib config
├── fp-lib-table             # Project-specific footprint lib config
├── lib/
│   ├── ProjectLib.kicad_sym  # Personal symbol library (this project's parts)
│   ├── ProjectLib.pretty/   # Personal footprint library (folder)
│   │   ├── MyChipPart.kicad_mod
│   │   └── CustomConnector.kicad_mod
│   └── 3d_models/
│       ├── MyChipPart.step
│       └── CustomConnector.wrl
├── docs/                   # Schematic PDF + photos
└── manufacturing/          # Gerbers + BOMs (auto-gen)
```

Git-Backed Library Workflow

- ▶ Symbol files (.kicad_sym) and footprint files (.kicad_mod) are TEXT
- ▶ Diff-friendly in Git — see exactly what changed between commits
- ▶ Tag library releases (v1.0, v1.1) for tracking
- ▶ Use Git submodules to share libraries between projects
- ▶ `git submodule add https://gitlab.com/myco/kicad-libs lib/CompanyLib`
- ▶ Updating company lib: `cd lib/CompanyLib && git pull`
- ▶ CI: run lint on library changes (KLC — KiCad Library Convention)
- ▶ Pull requests for library additions — code review = lib review

Corporate Library Best Practices

- ▶ Single source-of-truth Git repo
- ▶ Sub-libs by category (Passives, IC, Connector, etc.)
- ▶ Consistent naming: '<Vendor>_<MPN>_<Pkg>'
- ▶ Required fields: MPN, Datasheet URL, Manufacturer
- ▶ Required: 3D STEP model linked
- ▶ Required: review by 2 engineers before merge
- ▶ Versioning: semantic versions (1.0, 1.1, 2.0 = breaking)
- ▶ Changelog per release
- ▶ CI tests: KLC compliance, ERC test schematic
- ▶ Documentation per category
- ▶ Deprecation policy for obsolete parts
- ▶ Onboarding new engineers — point at the lib first

KLC — KiCad Library Convention

- ▶ KLC = official conventions for KiCad libraries
- ▶ Document at klc.kicad.org
- ▶ Rules for: symbol naming, footprint naming, pin numbers, 3D models
- ▶ Designed to ensure library consistency + interoperability
- ▶ Used by KiCad's built-in library (gold standard)
- ▶ Lint tool: `kicad-library-utils` (Python) checks compliance
- ▶ Useful for corporate library — adopt KLC as the standard
- ▶ Vendor libraries (SnapEDA) often NOT KLC-compliant — clean up before adding

Updating + Migration

- ▶ KiCad upgrade (7→8, 8→9) may require library format migration
- ▶ Run KiCad's library migrator (Tools → Manage Symbol/Footprint Libraries → Migrate)
- ▶ Backward compatibility: KiCad 9 reads KiCad 8 files; not always reverse
- ▶ Test old projects on new KiCad version before committing migration
- ▶ Tag library state in Git before migration (back-out path)
- ▶ Major rev (e.g., KiCad 7 → 8): allow 1-2 weeks of cohort migration time
- ▶ Communicate library changes to entire team

Common Library Mistakes

- ▶ Using built-in KiCad lib for vendor-specific parts → fab fails (wrong pinout)
- ▶ Single huge personal lib (1000+ parts) — hard to maintain
- ▶ No naming convention → duplicate parts proliferate
- ▶ Missing 3D models → mechanical engineers can't verify enclosure fit
- ▶ No MPN field → BOM export incomplete
- ▶ Outdated vendor lib → part EOL, no warning
- ▶ Library files not in Git → 'works on my machine' problem
- ▶ Adding parts without review → bad data propagates across projects

Plugin: Library Tools

kiCad-library-utils

Lint + check KLC compliance.
Open source.

SnapEDA Patrons

Auto-imports parts to lib.
Paid subscription.

Ultra Librarian sync

Auto-update vendor libs.

Library Loader (CSE)

Component Search Engine
plugin.

KiCad library scripts

Python scripts to clean up,
rename, merge libs.

Diff viewers

Visual diff of library changes
in Git.

Backup + Recovery

- ▶ Personal libs: in project Git → automatic backup
- ▶ Corporate libs: in shared Git repo → backup
- ▶ User's KiCad config (sym-lib-table, fp-lib-table): backup ~/.config/kicad
- ▶ If lib lost: re-clone from Git, run library migrator
- ▶ Use Git tags to mark fab releases — library state preserved with project
- ▶ Disaster: lib corruption → restore from Git tag, verify project still ERC clean
- ▶ Best practice: NEVER edit a lib outside Git — always commit changes

Onboarding Workflow

- ▶ Step 1: Install KiCad 8/9 (same version as team)
- ▶ Step 2: Clone corporate-lib Git repo to ~/kicad-lib/CompanyLib
- ▶ Step 3: Add to global libs: Tools → Manage Symbol/Footprint Libraries → Add
- ▶ Step 4: Configure environment variable KICAD_3DMODEL_DIR for 3D models
- ▶ Step 5: Clone any project repo with Git submodules → all libs come along
- ▶ Step 6: Run ERC test schematic with all corporate parts placed (verify libs OK)
- ▶ Step 7: Read KLC + corporate library doc before adding new parts

Key Takeaways

Key takeaways from this lecture

3-tier hierarchy

Project → global → built-in.

Personal libs per project

Vendor-specific parts.

Git-backed corporate lib

Shared, versioned.

KLC compliance

Industry standard convention.

3D models + MPN required

Mechanical + BOM.

Migrate on KiCad upgrade

Test before commit.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ klc.kicad.org — official Library Convention
- ▶ [kicad-library-utils](#) on GitHub — lint tool
- ▶ SnapEDA / Ultra Librarian / CSE for vendor parts
- ▶ KiCad docs: library management section
- ▶ Phil's Lab YouTube — library setup tutorials
- ▶ KiCad Forum — library best practices

Thank you • Build something this week.